SENSORS For Security Applications

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Electronic sensor systems extend their application to fields such as national defence, home, personal safety, office and industry

Are you familiar with the phrase "Big brother is watching you" from the novel *Big Brother-1984* by George Orwell? In the context of modern technology, the concept of watching could be compared to a CCTV system or other intelligent sensor system in security applications.

As technology progresses, threats and vulnerabilities to human lives including national security also increases. CCTV surveillance systems and other intelligent electronic sensors are being installed in many establishments, industrial setups and high-risk security areas to monitor and guard against these threats.

There are different types of electronic security systems used for the security of the nation, home, babies, the elderly, banks, cybernet, mobile and so on. Even farmers are facing security issues. It is being reported that bugs and weeds have become resistant to chemicals, resulting in the use of more pesticides and herbicides. This means, higher cost to farmers, more damage to environment and increasing health concerns.

Covering all security issues and sensors is out of the scope of this article. We have trimmed the article down to a few security applications such as national, home, personal, industrial and cybersecurity. Simple to complex sensor hardware and software are involved in guarding against the threats to human lives, properties, financial services and personal privacy.

Fig. 1: Representation of an intelligent electronic sensor (Credit: www.computerworld.com)

National security

There is a wide range of national security systems to guard against rogue states and terrorism, and monitor weather reports and natural calamities such as cyclones, earthquakes and tsunamis.

Fig. 2: Modern security systems (https://hdhtech.com)

A radar sensor is one of the most important electronics security systems. A radar transmits an electronic signal that bounces off objects and returns to the receiver for analysis. A radar sensor monitors areas such as national and international borders, military bases, airports, seaports, refineries and other critical industries. It detects movements at ground level, such as individuals walking or crawling towards a specific area, from a range of several kilometres. Radar sensor is also used in aircrafts, ships and submarines.

Satellite is another vital electronic system used in national security. Unlike radar, it is placed in an orbit above the Earth and uses cameras to take pictures of the Earth. Meteorologists use weather satellites and radars to monitor and forecast weather and atmospheric measurements, and provide important information on rain and snow, among others.

There are specialised electronic sensing equipment that are placed off shore and under seabeds to monitor earthquakes and tsunamis.

Recently, India and Israel held discussions regarding solutions to ensure security along international borders, using gadgets like electro-optical sensors and unmanned aerial vehicles (UAVs). Optical sensors like lidar (light detection and ranging) along with UAVs or drones could be used for efficient and reliable security applications. A lidar sensor mounted on a UAV along with